

ota-manager v1 router wiring + full tsc-0 remediation — EVIDENCE

Revision: 1 **Last modified:** 2026-07-09T21:09:04Z **Scope:** clients/ota-manager/ (Stream C-impl, §11.4.119 single-owner) **Authority:** operator decision “wire all v1 routes per design spec §4” **Anti-bluff:** §11.4 / §11.4.107 / §11.4.123 — every PASS below cites captured evidence; the one genuine backend-contract bug found (BUG-1) is reported, not hidden.

1. Summary verdict

Gate	Before	After	Evidence
tsc --noEmit	85 errors	0 errors (exit 0)	/tmp/ tsc_before.txt (85), live re-run exit 0
vitest run	—	36/36 pass (9 files)	live run, 6.13 s
vite build (harness)	—	success (1913 modules)	pnpm hostrender:build
Host-render — Login	PASS (prior)	PASS both themes	docs/qa/ 20260709-ota-manager-hostrender/
Host-render — Dashboard	(was placeholder)	PASS both themes	docs/qa/ 20260709-ota-manager-dashboard-hostrender/ results.json

Gate	Before	After	Evidence
v1 routes wired	1 (login-only effective)	9 routes (login + 8 layout children)	src/route-tree.gen.ts

All v1 feature pages are now wired into @tanstack/react-router. **No deferred sub-routes** — every list route AND both detail routes are mounted.

2. Routes wired (src/route-tree.gen.ts)

Code-based route tree, TanStack v1. layoutRoute is a pathless layout route (id: "layout", renders MainLayout = Sidebar + Topbar); all authenticated screens are its children (which is why useParams targets are prefixed /layout/...).

Path	Component	Kind
/login	LoginPage	root child (unauthenticated)
/dashboard	DashboardPage (feature-rich)	layout child
/devices	DevicesPage	layout child (list)
/devices/\$deviceId	DeviceDetailPage	layout child (detail)
/releases	ReleasesPage	layout child (list)
/deployments	DeploymentsPage	layout child (list)
/deployments/\$deploymentId	DeploymentDetailPage	layout child (detail)
/groups	GroupsPage	layout child (list)
/audit	AuditPage	layout child (list)

Before this work the shipped route tree effectively mounted only / login + a static placeholder /dashboard; the rich feature pages existed in src/features/** but were **unwired dead code**. They are now the real routed screens.

3. The 85 tsc errors — categorised fixes

Baseline breakdown by TS code (/tmp/tsc_before.txt, 85 total):

TS code	Count	Meaning	Category
TS2339	38	property does not exist	hook-return-shape drift + API model drift
TS2345	13	argument type mismatch	filter / input-shape drift
TS2322	11	type not assignable	view-model / prop drift
TS6133 / TS6192	6 + 2	unused local / all-imports unused	dead imports
TS2353 / TS2561	5 + 1	unknown object property	Toast / Select / form props
TS2724 / TS2305	2 + 1	wrong / missing exported member	router API migration
TS2554	2	wrong argument count	hook signature drift
TS2352	2	unsafe cast	view-model cast
TS7031 / TS2740	1 + 1	implicit any / missing properties	form typing

Category A — router-API migration (react-router-dom → @tanstack/react-router)

- src/features/dashboard/dashboard-page.tsx — useNavigate now from @tanstack/react-router; quick-action handlers use navigate({ to: "/releases" | "/deployments" | "/devices" | "/groups" }).
- src/features/layout/sidebar.tsx — replaced the invalid className={{({ isActive }) => ...} render-prop (react-router-dom API) with TanStack Link activeProps / inactiveProps.

- `src/features/deployments/deployment-detail-page.tsx` — `useParams({ from: "/layout/deployments/$deploymentId" })`.
- `src/features/devices/device-detail-page.tsx` — `useParams({ from: "/layout/devices/$deviceId" })`.
- `src/__tests__/dashboard.test.tsx` — mock switched from `react-router-dom` to `@tanstack/react-router`.

Root-cause note (TanStack layout-route ID prefix): `useParams` on a child of a pathless layout route (`id: "layout"`) requires the layout-prefixed route id `"/layout/.../$param"`, not the bare path. Discovered as the last 2 residual errors and fixed → tsc 0.

Category B — hook-return-shape drift (adapter-shim pattern)

The `src/features/**` pages consume camelCase *view models* (`{ devices, loading, error, refresh }`, `Release`, `Deployment`, `Group`, `AuditView`, ...) while the underlying react-query hooks return raw wire `*List` bodies. Rather than rewrite every page (and break the mocked tests), the camelCase hook files were rewritten as **adapters** that map `wire` → `view shape` and re-export the underlying query hooks: `useDevices.ts`, `useReleases.ts`, `useDeployments.ts`, `useGroups.ts`, `useAuditLog.ts`, `useDeployment.ts`, `useCreateDeployment.ts`, `useCreateGroup.ts`, `useCreateRelease.ts`, `useEvaluateRollout.ts`, `useRecall.ts`, `useArtifact.ts`, `useUploadArtifact.ts`. Each adapter's field mapping was verified against the **real Go server wire contract** (`server/internal/api/*_wire.go / handlers`) — e.g. `Release.os` ← `firmware_version`, `Release.targetModel` ← `target_board`, `Deployment.id` ← `deployment_id`, `Group.memberCount` ← `device_count`, `AuditView.timestamp` ← `created_at`, `target` ← `resource_type/resource_id`.

Category C — component / form / toast props

- `src/components/ui/select.tsx` — `SelectValue` gained a placeholder prop.
- `src/components/ui/form.tsx` — `Form` = `FormProvider`; `FormField` wraps `react-hook-form` `Controller` with correct `Control/FieldValues/FieldPath` generics.
- `src/hooks/use-toast.ts` — exposes `toast()` mapping to context `addToast`.

Category D — unused imports / dead code

- Removed unused imports in `devices-page.tsx`, `releases-page.tsx` (`CardDescription`), `groups-page.tsx` (`SheetTrigger`), `deployments-page.tsx`, `deployment-detail-page.tsx` (`Dialog block`, `page-level useToast`), `device-detail-page.tsx` (`React`).

4. §11.4.124 investigate-before-remove — deletions (all removal-proof captured)

Three files were removed. Each was verified **zero-importer** before removal (grep of the importer graph + test inspection) and each removal is preserved in git history:

File	Reason	Git evidence
<code>src/routes/index.tsx</code>	file-based routing API (10 errors), superseded by code-based route-tree.gen.ts; zero importers	committed 9d21dd6a (“3 files changed, 332 deletions”)
<code>src/routes/dashboard.tsx</code>	static data-free placeholder, replaced by the wired feature-rich features/dashboard/dashboard-page.tsx; zero importers	committed 9d21dd6a
<code>src/features/layout/app-layout.tsx</code>	dead duplicate of <code>MainLayout</code> (3 errors), never referenced by the route tree; zero importers	committed 9d21dd6a

Verification (this session): all three are ABSENT-ON-DISK, NOT-IN-INDEX, NOT-IN-HEAD — the removals persisted cleanly.

5. Host-render dual-oracle proofs (§11.4.170)

Both harnesses mount the **real, unmodified** `route-tree.gen.ts` and render the real component to PNG on the host via Playwright, with three self-validated analyzers (image-diff, OCR, layout) each carrying a golden-good (must PASS) + golden-bad (must FAIL) pair.

5.1 Login — PASS both themes

`docs/qa/20260709-ota-manager-hostrender/` · fresh run this session: - image-diff golden-good 0.0000% (PASS) / golden-bad ~1.5% (FLAGGED) — both themes - OCR all labels present; golden-bad correctly flags missing “OTA Manager” - layout OK; golden-bad flags collapsed submit button - analyzer self-validation: image-diff / layout / OCR all sound: true

5.2 Dashboard — PASS both themes

`docs/qa/20260709-ota-manager-dashboard-hostrender/results.json` · run this session (`visual/run-all-dashboard.mjs`): - image-diff golden-good 0.0000% (PASS) / golden-bad 2.97% light, 3.14% dark (FLAGGED) - OCR all labels present (Dashboard, Total Devices, Active Deployments, Pending Updates, Failed Devices, Recent Activity, Quick Actions); golden-bad flags missing “Total Devices” - layout OK (no collapse/clip/offscreen/overlap); golden-bad flags collapsed title - analyzer self-validation: all three sound: true

Scope of the Dashboard claim (anti-bluff, §11.4.107 / §11.4.123): the harness feeds `DashboardPage` its **declared TypeScript input contract** — the `camelCase TelemetryOverview` view-model the component consumes — via a Playwright-layer `route. fulfill stub` for `GET /telemetry/overview` (+ empty `GET /audit`). It therefore proves the **screen renders correctly for that view-model** in both themes with all oracles green. It does **NOT** assert end-to-end backend integration — see BUG-1.

5.3 Other newly-wired data screens — host-render SKIP-with-reason (honest)

`/devices, /devices/$id, /releases, /deployments, /deployments/$id, /groups, /audit` were **not** given dedicated §11.4.170 host-render harnesses in this session. - **Reason:** each requires a bespoke Playwright network-stub harness per screen×state; that is a separate, sizeable

work item (one harness entry + runner + golden-good/golden-bad fixtures per screen). Building them was outside this session's tsc-0 + route-wiring budget. - **What DOES cover them now:** (a) tsc --noEmit 0 (type-level correctness of every page against the adapter view-models), (b) vitest 36/36 (component render + interaction unit tests with mocked hooks), (c) the shared render pipeline is proven live by the Login + Dashboard host-render runs. This is an honest SKIP, not a faked PASS.

6. BUG-1 (real bug found, reported not hidden) — telemetry-overview client/server model drift

Severity: functional (dashboard stat cards read as zeros against the real server). **Class:** client/server wire-contract drift. **Pre-existing** — not introduced by this router-wiring work; surfaced by it. Does **not** produce a tsc error (both sides are internally type-consistent; the mismatch is runtime).

- Client TelemetryOverview (src/lib/api-client.ts:180):
{ totalDevices, activeDeployments, pendingUpdates, failedDevices } (camelCase).
- useTelemetryOverview (src/hooks/use-devices.ts:92) casts the raw / telemetry/overview body to that type with **no mapping**.
- Real Go server (server/internal/api/handlers_telemetry.go:56) returns { event_counts, total, failure_rate, by_state } — **none of the four camelCase fields exist**.
- Effect: overview?.totalDevices etc. are undefined ?? 0 → all four stat cards show **0** against the real server.

Why not fixed here: a correct fix is a *product-design* change, not a rename — the server's overview endpoint does not expose an “active deployments” count at all, and the other three stats would have to be derived from by_state / event_counts (which of by_state's keys map to “pending updates” / “failed devices” is a product decision). That is out of scope for “wire the routes + drive tsc to 0” and is owned by whoever owns the telemetry contract (server + client-model). **Reported for the conductor/operator to route.** The Dashboard host-render proof is scoped accordingly (§5.2) and makes no end-to-end claim.

KNOWN GAPS (also honest, no bluff): the server exposes **no list-artifacts endpoint**, so useUploadArtifact returns an empty artifact-option list and the create-release artifact picker is submission-

incomplete. GroupDetailContent member management is a documented placeholder (the membership hooks exist; wiring the member table is deferred).

7. Full file inventory

Modified (uncommitted working-tree delta, this session's ownership)

```
src/route-tree.gen.ts
src/components/ui/select.tsx
src/components/ui/form.tsx
src/hooks/use-toast.ts
src/hooks/useDevices.ts
src/hooks/useReleases.ts
src/hooks/useDeployments.ts
src/hooks/useDeployment.ts
src/hooks/useGroups.ts
src/hooks/useAuditLog.ts
src/hooks/useCreateDeployment.ts
src/hooks/useCreateGroup.ts
src/hooks/useCreateRelease.ts
src/hooks/useEvaluateRollout.ts
src/hooks/useRecall.ts
src/hooks/useArtifact.ts
src/hooks/useUploadArtifact.ts
src/features/dashboard/dashboard-page.tsx
src/features/layout/sidebar.tsx
src/features/devices/devices-page.tsx
src/features/devices/device-detail-page.tsx
src/features/releases/releases-page.tsx
src/features/deployments/deployments-page.tsx
src/features/deployments/deployment-detail-page.tsx
src/features/groups/groups-page.tsx
src/__tests__/dashboard.test.tsx
visual/run-all-dashboard.mjs          (harness re-pointed from
retired placeholder to the wired rich dashboard + telemetry/audit
stubs)
```

Deleted (already committed 9d21dd6a; §11.4.124 evidence in §4)

```
src/routes/index.tsx
src/routes/dashboard.tsx
src/features/layout/app-layout.tsx
```

Created

```
docs/qa/20260710-ota-manager-router-wiring/EVIDENCE.md    (this
file)
```

(Harness scaffolding `visual/harness-dashboard.{html,tsx}` and the `harness-dashboard` build input were added earlier in this wave and are already tracked; regenerated `dist/**` and `visual/.out/**` are build artifacts, not hand edits — left for the conductor to handle per its commit policy.)

8. Reproduce

```
cd clients/ota-manager
pnpm exec tsc --noEmit           # exit 0, no output
pnpm vitest run                  # 36/36 pass
pnpm hostrender:build            # vite build, 1913 modules
node visual/run-all.mjs         # Login host-render — OVERALL:
    PASS
node visual/run-all-dashboard.mjs # Dashboard host-render —
    OVERALL: PASS
```